OPERATIONS MANAGEMENT

UNIT – I I

FACILITY/PLANT LOCATION/ PLANT LAYOUT

Facility /Plant Location

Facility location may be defined as a place where the facility will be set up for producing goods or services. The need for location selection may arise under any of the following conditions:

- When a business is newly started.
- When the existing business unit has outgrown its original facilities and expansion is not possible; hence a new location has to be found.
- When the volume of business or the extent of market necessitates the establishment of branches.
- When the lease expires and the landlord does not renew the lease.
- Other social or economic reasons.

Plant location or the facilities location problem is an important strategic level decision- making for an organisation. One of the key features of a conversion process (manufacturing system) is the efficiency with which the products (services) are transferred to the customers. This fact will include the determination of where to place the plant or facility.

The selection of location is a key-decision as large investment is made in building plant and machinery. It is not advisable or not possible to change the location very often. So, an improper location of plant may lead to waste of all the investments made in building and machinery, equipment.

Before a location for a plant is selected, long range forecasts should be made anticipating future needs of the company. The plant location should be based on the company's expansion plan and policy, diversification plan for the products, changing market conditions, the changing sources of raw materials and many other factors that influence the choice of the location decision. The purpose of the location study is to find an optimum location one that will result in the greatest advantage to the organization.

Objectives of plant location:

- Reduced capital investment and operating cost
- Ensuring effective plant layout
- Coordination with government policies
- Employee welfare and public needs Security

Aspects of plant location:

There are three aspects;

1. Selection of a region
2. Selection of a locality (community factor)
3. Selection of a site

Need for location selection:

I. When starting a new organization, i.e., location choice for the first time.

II. In case of existing organization.

III. In case of Global Location.

I. In case of location choice for the first time or new organizations:

Cost economies are always important while selecting a location for the first time, but should keep in mind the cost of long-term business/organisational objectives. The following are the factors to be considered while selecting the location for the new organizations:

Identification of region: The organisational objectives along with the various long-term considerations about marketing, technology, internal organisational strengths and weaknesses, region specific resources and business environment, legal-governmental environment, social environment and geographical environment suggest a suitable region for locating the operations facility.

Choice of a site within a region: Once the suitable region is identified, the next step is choosing the best site from an available set. Choice of a site is less dependent on the organisations long-term strategies. Evaluation of alternative sites for their tangible and intangible costs will resolve facilities-location problem. The problem of location of a site within the region can be approached with the following cost-oriented noninteractive model, *i.e.*, dimensional analysis.

II. In case of location choice for existing organization:

In this case a manufacturing plant has to fit into a multi-plant operations strategy. That is, additional plant location in the same premises and elsewhere under following circumstances:

- Plant manufacturing distinct products.
- Manufacturing plant supplying to specific market area.
- Plant divided on the basis of the process or stages in manufacturing.
- Plants emphasizing flexibility.

III. In case of global location:

Because of globalization, multinational corporations are setting up their organizations in India and Indian companies are extending their operations in other countries. In case of global locations there is scope for virtual proximity and virtual factory.

Virtual Proximity: With the advance in telecommunications technology, a firm can be in virtual proximity to its customers. For a software services firm much of its logistics is through the information/ communication pathway. Many firms use the communications highway for conducting a large portion of

their business transactions. Logistics is certainly an important factor in deciding on a location whether in the home country or abroad. Markets have to be reached. Customers have to be contacted. Hence, a market presence in the country of the customers is quite necessary.

Virtual Factory: Many firms based in USA and UK in the service sector and in the manufacturing sector often out sources part of their business processes to foreign locations such as India. Thus, instead of one's own operations, a firm could use its business associates operations facilities.

Factors Affecting Facility Location Decisions

While selecting a facility location, an organisation should consider various factors that may have significant impact on its performance. These factors are explained below:

- Availability of Raw Materials
- Availability of power
- Transportation
- Suitability of climate
- Government policy
- Competition between states

Availability of raw materials:

It is essential for the organization to get raw material in right qualities and time in order to have an uninterrupted production. The cost obtaining Example Nearness to raw material is important in case of industries such as sugar, cement, jute and cotton textiles.

Nearness to market:

Every company is expected to serve its customers by providing goods and services at the time needed and at reasonable price organizations may choose to locate facilities close to the market or away from the market depending upon the product. When the buyers for the product are concentrated, it is advisable to locate the facilities close to the market.

Transportation facilities:

Speedy transport facilities ensure timely supply of raw materials to the company and finished goods to the customers. The transport facility is a prerequisite for the location of the plant. There are five basic modes of physical transportation, air, road, rail, water and pipeline. Goods that are mainly intended for exports demand a location near to the porter large airport. The choice of transport method and hence the location will depend on relative costs, convenience, and suitability. Thus, transportation cost to value added is one of the criteria for plant location.

Climatic conditions: The geology of the area needs to be considered together with climatic conditions (humidity, temperature). Climates greatly influence human efficiency and behaviour. Some industries require specific climatic conditions e.g., textile mill will require humidity.

Government policy:

The policies of the state governments and local bodies concerning labour laws, building codes, safety, etc., are the factors that demand attention. In order to have a balanced regional growth of industries, both central and state governments in our country offer the package of incentives to entrepreneurs in particular locations. The incentive package may be in the form of exemption from a sales tax and excise duties for a specific period, soft loan from financial institutions, subsidy in electricity charges and investment subsidy. Some of these incentives may tempt to locate the plant to avail these facilities offered.

Labour and wages:

The problem of securing adequate number of labour and with skills specific is a factor to be considered both at territorial as well as at community level during plant location. Importing labour is usually costly and involve administrative problem. Productivity of labour is also an important factor to be considered. Prevailing wage pattern, cost of living and industrial relation and bargaining power of the union's forms in important considerations.

Rural v/s Urban Sites

S.No	Urban	Rural
1	Urban areas usually refer to cities, suburbs and towns.	Rural areas usually refer to villages
2	Urban areas have more development in terms of access to infrastructure and connectivity like airports, ports, railways, housing, roads etc.	Rural areas usually don't have much development in terms of infrastructure.
3	Land in urban areas is used for development activities. There is usually not much land available which has not been used for developmental activities	Rural areas usually have a lot of vacant lands without much development.
4	Urban areas are densely populated	Rural areas are sparsely populated
5	As of 2018, approximately 34% of India's population lives in urban areas	More than 65% of India's population lives in rural areas as per 2018 figures. From 2008 to 2018 there has been a steady decline in the percentage of the population living in rural areas of India, due to migration.
6	Although the total population living in urban areas is much less compared to	The population density is less in rural areas compared to urban areas.

	Rural areas, the population density is very high in urban areas	
7	Urban areas have a scarcity of land	Rural areas do not have land scarcity.

Facility/plant layout

After the site location decision has been made, the next focus in production planning is the facility's layout. The goal is to determine the most efficient and effective design for the particular production process.

Plant Layout Definition

According to F G. Moore, "A good layout is one which allows materials rapidly and directly for processing. This reduces transport handling, clerical and other costs down per unit, space requirements arc minimized and it reduces idle machine and idle man time."

"Plant layout is a plan for effective utilisation of facilities for the manufacture of products; involving a most efficient and economical arrangement of machines, materials, personnel, storage space and all supporting services, within available floor space".

"Plant layout is a plan of optimum arrangement of facilities including personnel, equipment's, storage space, material handling equipment and all other supporting services along with the decision of best structure to contain all these facilities."

Needs for plant layout

- When a new plant is set up to manufacture newly designed products.
- When a new product line is introduced in the factory.
- When additional space is required to meet increased demand for products.
- When frequent accident occurs in the factory.
- When the existing facilities are to be rearrangement due to change in methods or production.
- When the existing factory is shifted to a new area.
- When the cost reduction program is launched.
- When improvements are to be made by incorporating scientific and technological innovations.

Objectives of Facility Layouts

Objectives for Manufacturing Operation Layouts

- Provide enough production capacity
- Reduce materials-handling costs
- Conform to site and building constraints
- Allow space for production machines
- Allow high labor, machine, and space utilization and productivity
- Provide for volume and product flexibility

- Provide space for rest rooms, cafeterias, and other personal-care needs of employees
- Provide for employee safety and health
- Allow ease of supervision
- Allow ease of maintenance
- Achieve objectives with least capital investment

Additional Objectives for Warehouse Operation Layouts

- Promote efficient loading and unloading of shipping vehicles
- Provide for effective stock picking, order filling, and unit loading
- Allow ease of inventory counts
- Promote accurate inventory record keeping

Additional Objectives for Service Operation Layouts

- Provide for customer comfort and convenience
- Provide appealing setting for customers
- Allow attractive display of merchandise
- Reduce travel of personnel or customers Provide for privacy in work areas
- Promote communication between work areas
- Provide for stock rotation for shelf life

Additional Objectives for Office Operation Layouts

- Reinforce organization structure
- Reduce travel of personnel or customers
- Provide for privacy in work areas
- Promote communication between work areas

Factors affecting the plant layout decision

1. **Type of production**: - the plant layout for engineering unit will be quite different from that of a flour factory.
2. **Production system:** - the plant layout is a continuous production system will be totally different that under the intermitted production system.
3. **Scale of production: -** the plant layout and material handling equipment in the large scale organization will be different from that in the small-scale manufacturing activity.
4. **Type of machine: -** the use of simple-purpose and multipurpose machines substantially affect the plant layout. Similarly noisy and vibrating machines require special attention in the plant layout decision.
5. **Type of building facility: -** the plant layout in a single-story building will be different from that in a multi-story building.

6. **availability of total floor area: -** the allocation of space for machines, work benches, etc is made on the basis of the available floor area. Use of overhead space is made in case of shortage of space.
7. **possibility of future expansion: -** the plant layout is made in the light of the future requirements and installation of additional facilities.
8. **arrangement of material handling equipment:** - the plant layout and the material handling services are closely related and the latter has a decisive effect on the arrangement of production process and plant services.

Determinants of Layout

Although our introduction to layouts comes in the context of new locations, any significant change in operations may warrant a review (and possibly a revision) of an existing layout. The type of layout is generally determined by the following:

Type of product: This concerns whether the product is a good or a service, the product design and quality standards, and whether the product is produced for stock or for order.

Type of production process: This relates to the technology used, the type of materials handled, and/or the means of providing the service.

Volume of production: Volume affects the present facility design and capacity utilization, plus provisions for expansion or change.

Types of Plant Layout:

Layouts can be classified into the following five categories:

1. Product layout
2. Process layout
3. Combination layout
4. Fixed position layout
5. Service Facility layout

Product Layout (or Line Layout):

In this type of layout, all the machines are arranged in the sequence, as required to produce a specific product. It is called line layout because machines are arranged in a straight line. The raw materials are fed at one end and taken out as finished product to the other end. The raw material is supplied at one end of the line and goes from one operation to the next quite rapidly with a minimum work in process, storage and material handling.

Advantages offered by Product Layout:

(i) Lowers total material handling cost.
(ii) There is less work in processes.
(iii) Better utilization of men and machines,
(iv) Less floor area is occupied by material in transit and for temporary storages.

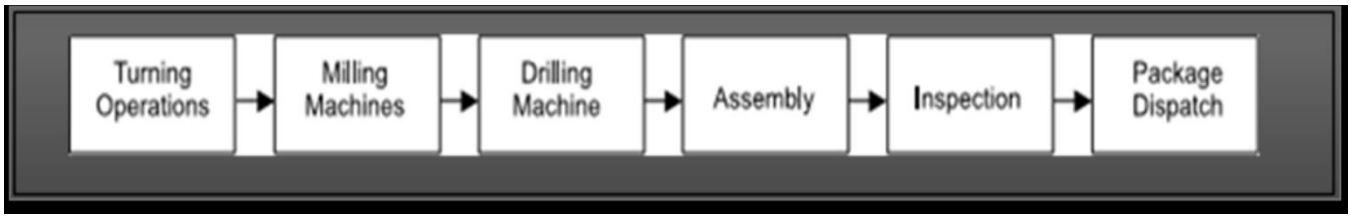
(v) Greater simplicity of production control.

(vi) Total production time is also minimized.

Limitations of Product Layout:

i. No flexibility which is generally required is obtained in this layout.

ii. The manufacturing cost increases with a fall in volume of production.



Process or Functional Layout:

The process layout is particularly useful where low volume of production is needed. If the products are not standardized, the process layout is lower desirable, because it has creator process flexibility than other. In this type of layout, the machines and not arranged according to the sequence of operations but are arranged according to the nature or type of the operations. This layout is commonly suitable for non-repetitive jobs. Same type of operation facilities is grouped together such as lathes will be placed at one place; all the drill machines are at another place and so on. Therefore, the process carried out in that area is according to the machine available in that area.

Advantages of Process Layout:

(i) There will be less duplication of machines. Thus, total investment in equipment purchase will be reduced.

(ii) It offers better and more efficient supervision through specialization at various levels.

(iii) There is a greater flexibility in equipment and man power thus load distribution is easily controlled.

(iv) Better utilization of equipment available is possible.

(v) Break down of equipment can be easily handled by transferring work to another machine/work station.

(vi) There will be better control of complicated or precision processes, especially where much inspection is required.

Limitations of Process Layout:

(i) There are long material flow lines and hence the expensive handling is required.

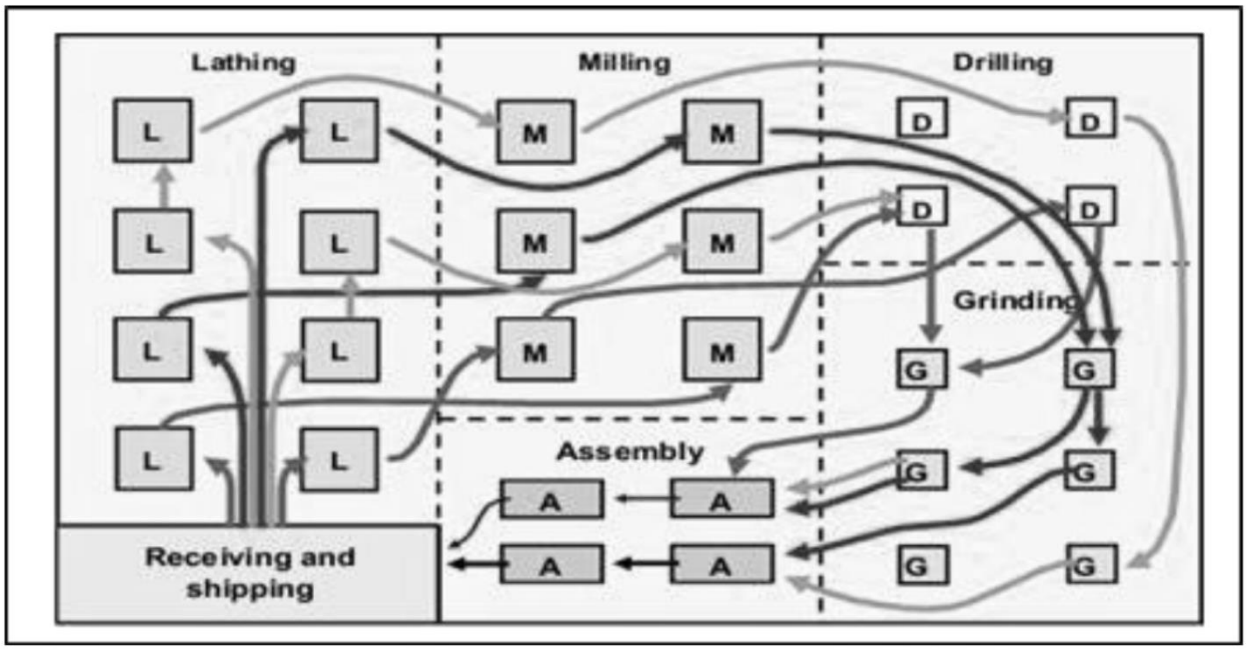
(ii) Total production cycle time is more owing to long distances and waiting at various points.

(iii) Since more work is in queue and waiting for further operation hence bottle necks occur.

(iv) Generally, more floor area is required.

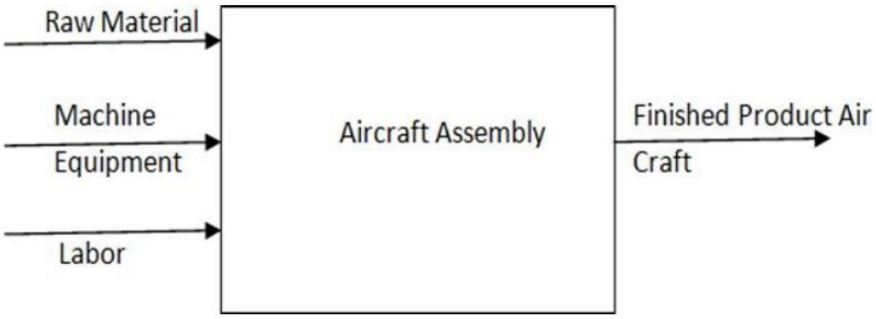
(v) Since work does not flow through definite lines, counting and scheduling is more consuming.

(vi) Specialization creates monotony and there will be difficult for the laid workers to find job in other industries.



Fixed Position Layout:

This type of layout is the least important for today's manufacturing industries. In this type of layout, the major component remains in a fixed location, other materials, parts, tools, machinery, man power and other supporting equipment's are brought to this location. The major component or body of the product remain in a fixed position because it is too heavy or too big and as such it is economical and convenient to bring the necessary tools and equipment's to work place along with the man power. This type of layout is used in the manufacture of boilers, hydraulic and steam turbines and ships etc.



Advantages of Fixed Position Layout:

(i) Material movement is reduced.

(ii) Capital investment is minimized.

(iii) The task is usually done by gang of operators, hence continuity of operations is ensured.

(iv) Production centers are independent of each other. Hence, effective planning and loading can be made. Thus, total production cost will be reduced.

(v) It offers greater flexibility and allows change in product design, product mix and production volume.

Limitations of Fixed Position Layout:

(i) Highly skilled man power is required.

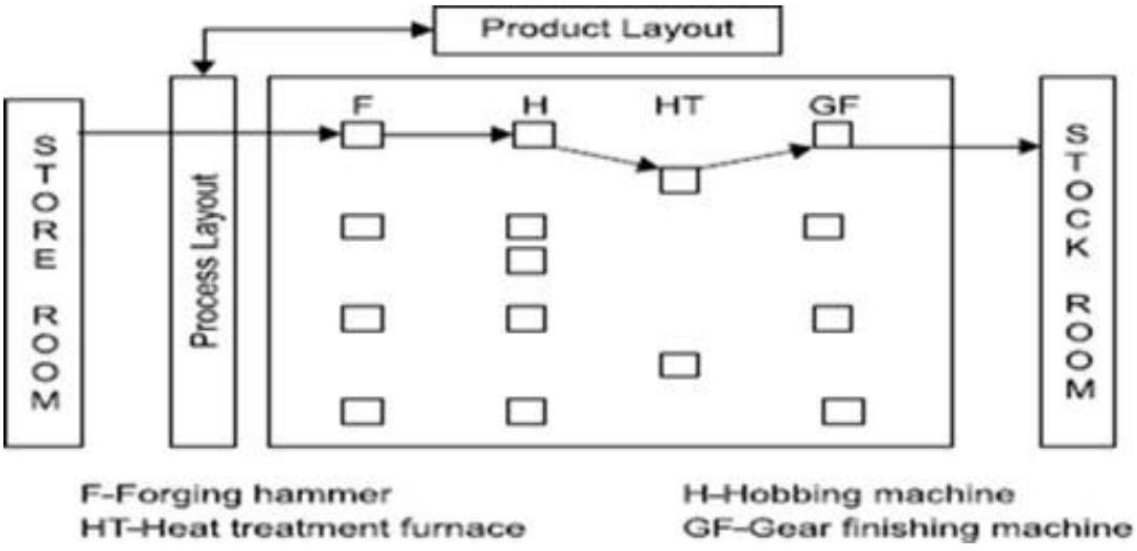
(ii) Movement of machines equipment's to production centre may be time consuming.

(iii) Complicated fixtures may be required for positioning of jobs and tools. This may increase the cost of production.

Combination Type of Layout:

Now a day in pure state any one form of layouts discussed above is rarely found. Therefore, generally the layouts used in industries are the compromise of the above-mentioned layouts. Every layout has got certain advantages and limitations. Therefore, industries would to like use any type of layout as such. Flexibility is a very important factory, so layout should be such which can be molded according to the requirements of industry, without much investment. If the good features of all types of layouts are connected, a compromise solution can be obtained which will be more economical and flexible.

In practice, plants are rarely laid out either in product or process layout form. Generally a combination of the two basic layouts is employed; to derive the advantages of both systems of layout. For example, refrigerator manufacturing uses a combination layout.



Advantages:

(i) Component standardization rationalization.

(ii) Effective machine operation and productivity.

(iii)Customer service.

(iv)It can decrease the paper work and overall production time.

(v) Work-in-progress and work movement

Service Facility Layout:

Services facility layouts should provide for easy entrance to these facilities Large, well organized and well-designed walk ways to and from parking areas are some of the requirements of service requirements The fundamental difference between service facility and manufacturing facility layouts is that many service facilities exist together customers and services.

Examples; Hospitals – processing of physical materials and production efficiency. Banks - customer receiving and servicing function.

Comparison of Layouts for Goods and Services

Layouts do not usually conform to the classifications described above in all respects. For example, behind the picturesque, smooth-running automobile assembly lines lie job and batch processing shops producing the needed components and subassemblies. And some of these production activities are highly automated whereas others are not.

Service systems also blend characteristics of job shop, batch, and line processing layouts. At one end of the spectrum, hospitals make full use of the job shop characteristics. Patients (instead of materials) are uniquely routed to different "work centers" for customized x-rays, EKGs, blood tests, surgery, and so forth. Like machine shops, many service facilities also rely on queues to regulate the flow of customers. However, empty hospital beds, idle firefighters, and unfilled airline seats represent the costs of idle capacity, much as idle machines do in a manufacturing plant.

But as the services offered become more standardized, characteristics of continuous flow emerge. Customers entering movie theaters and subway systems are "batch processed" in similar ways by personnel and equipment geared to a limited variety of outputs.

Line processing and continuous flow service systems also capitalize on the same economies of scale (high volume and low processing costs) as manufacturing systems. And as the service becomes highly standardized, the "tailored" nature of the service (or the marketing strategy) shifts to choices concerning the time and quantity of service desired. The use of electricity, phone service, stock quotations, computer banking, and so forth illustrate this shift. An increasing number of these continuous flow-type services are information- and/or communications-oriented. So, the "layout strategies" here focus on better ways of making the service available wherever and whenever the customer wants it.